Draft: game-built quadratic forms in the nimber Clifford backend

a9lim

June 2026

Abstract

This is a consolidated experimental note about the part of `ogdoad` that seems mathematically worth isolating. Conway games under disjunctive sum do not form a scalar ring, so the project does not construct "Clifford algebras over all games". Instead it builds Clifford algebras over field-like game-value worlds, with emphasis on the characteristic-2 nimber backend, whose engine keeps the quadratic data $q_i = e_i^2$ and the polar data $b_{ij} = e_i e_j + e_j e_i$ independent and computes trace-Arf invariants of quadratic forms over finite nimber subfields.

The experimental bridge is this: the Gold trace forms

$$Q_a(x) = \mathrm{Tr}(x\, x^{2^a})$$

are standard quadratic forms over fields of characteristic 2, but in the nimber setting they are also composites of combinatorial-game operations: nim-product via Turning Corners, Frobenius squaring via the diagonal product, and XOR via disjunctive sum. Their Arf invariant has the classical zero-count interpretation, so if a natural game had P-positions exactly $\{x : Q_a(x) = 0\}$, Arf would be the sign of the second-player win-bias. The note does not claim such a game has been found.

Beyond the construction, the note synthesizes three threads of work on the open game-semantics question. First, the obstruction landscape: normal-play sums and Winning-Ways coin-turning realize only *linear* P-sets (the lexicode phenomenon is the rich solved case of that linear ceiling), and a family of no-go theorems eliminates symmetric, commutative, oracle-bounded, and translation-equivariant architectures above it. Second, an extraspecial-group reframing: a characteristic-2 quadratic form *is* a central extension $1 \to \mathbb{Z}/2 \to E \to V \to 1$ whose squaring map is Q, which forces the quadratic datum to come from a noncommutative source and yields a naturality screen strictly between the frame-blind no-go and tautological evaluation. Third, the constructive boundary: within every tested bounded-access class, exact realizers of $\{Q = 0\}$ exist but are forced clocks; one charge-counting construction (ECHO-ko) is exact on the verified rank-4 bent instances and near-exact at low rank, while remaining decision-live. A formalized naturality criterion makes the residual question precise: does a decision-nondegenerate fixed uniform rule realize a Gold quadric?

Claim levels. Following the repository convention, statements below are labelled *standard math* (external facts with citations), *implemented and tested* (backed by tests, examples, or experiments in this checkout), *interpretation* (conditional bridges), or *open*. Several results of the 2026-06-10 parallel research run are machine-verified only at stated small parameters; each such result carries its verification scope inline. One construction claim (ECHO-FIFO+dummy, Section 10), recorded as unverified in earlier drafts, has since passed its pre-registered adversarial review (2026-06-10): a fresh direct stateful solver (`experiments/echo_solver.py`) re-derives the full 391,680-check $m = 8$ exactness sweep with zero misses. The verification record is in Section 10.

1 Scope

A short partizan game is built recursively as $\{L \mid R\}$, and games form a partially ordered abelian group under disjunctive sum [4, 3]. They do not form a ring: Conway multiplication is defined on the number subclass, not on arbitrary games. Since a Clifford algebra needs a commutative scalar ring, the phrase "Clifford over games" has to be interpreted through scalar subworlds:

world	algebraic structure	role here
No	real-closed field, char 0	ordinary real Clifford theory with surreal scalars
No$[i]$	algebraically closed field, char 0	ordinary complex Clifford theory
finite nimber subfields	finite fields of char 2	characteristic-2 Clifford and Arf experiments

The implemented `Nimber` scalar is finite: `u128` realizes $\mathbb{F}_{2^{128}}$, containing the nimber subfields $\mathbb{F}_{2^m}$ for $m = 1, 2, 4, \ldots, 128$. It is not the whole proper-class field On_2.

2 The char-2 Clifford datum

In characteristic 2, a quadratic form is not determined by its polar form. For basis vectors the Clifford relations are

$$e_i^2 = q_i, \qquad e_ie_j + e_je_i = b_{ij}.$$

The polar form is alternating, so $b_{ii} = 0$, but q_i can be nonzero. A faithful characteristic-2 implementation must store q and b separately. This is the main implementation discipline behind the nimber backend, and – as Section 7 makes precise – it is also the group-theoretic content of the whole note: q and b are the independent central data (squares and commutators) of an extraspecial-type extension.

For a nonsingular quadratic form over $\mathbb{F}_2$ with symplectic basis (a_k, b_k), the Arf invariant is

$$\mathrm{Arf}(Q) = \sum_k Q(a_k)Q(b_k) \in \mathbb{F}_2.$$

It classifies nonsingular binary quadratic spaces at fixed dimension. Over a finite characteristic-2 field, the corresponding value lies in $F/\wp(F)$, with $\wp(y) = y^2 + y$; for finite fields this quotient is read by the field trace to $\mathbb{F}_2$.

Remark 1. Ovsienko [11] identifies a precise bridge between Dickson's classification of quadratic forms over $\mathbb{F}_2$ and the classical classification of real Clifford algebras. We use that paper as motivation for the Arf/Clifford connection, not as a blanket classification theorem for all Clifford algebras over finite nimber fields. In the code, rank and radical data are reported alongside Arf, and degenerate forms are not collapsed to a single Arf bit.

The small finite-field check used throughout is:

$$q = (*2, *2),\ b_{01} = *1 \quad \text{has Arf } 1,$$

whereas

$$q = (*2, *3),\ b_{01} = *1 \quad \text{has Arf } 0.$$

The polar entry is part of the form.

3 Gold forms from game operations

Nim-addition is XOR, and XOR is the disjunctive sum of impartial game values. Nim-multiplication is also game-theoretic: Conway's Turning-Corners product has the mex recurrence

$$x \otimes y = \operatorname{mex}\{(i \otimes y) \oplus (x \otimes j) \oplus (i \otimes j) : 0 \le i < x,\ 0 \le j < y\}. \tag{1}$$

The repo implements this recurrence directly for small arguments and checks it against the fast algebraic nim product.

Thus the following operations are game-realizable at the level of nimber values:

$$x \oplus y, \qquad x \otimes y, \qquad x \mapsto x^2 = x \otimes x, \qquad \operatorname{Tr}(x) = x + x^2 + \cdots + x^{2^{m-1}}.$$

For $F = \mathbb{F}_{2^m}$ and $a \ge 1$, define the Gold trace form

$$Q_a(x) = \operatorname{Tr}\big(x\, x^{2^a}\big) = \operatorname{Tr}\big(x^{1+2^a}\big).$$

Since Frobenius $x \mapsto x^{2^a}$ is additive, Q_a is a quadratic form over $\mathbb{F}_2$. In the nimber backend, it is also a composite of the game operations above. Throughout, "the instance (m, a)" means Q_a on $\mathbb{F}_{2^m}$, and "(m, a, λ)" the scaled component $Q_{\lambda,a}(x) = \operatorname{Tr}(\lambda x^{1+2^a})$.

Observation 1. *The Gold forms are not merely defined on a field of game values. In the finite nimber fields tested here, they are built from nim/game operations: Turning Corners for multiplication, diagonal Turning Corners for Frobenius squaring, and disjunctive sum for XOR and trace.*

This observation is modest but useful: it gives a concrete quadratic form at the interface between impartial-game arithmetic and characteristic-2 quadratic-form theory.

3.1 Rank check

The polar rank of Gold forms is known. In the tested regime $m = 2^k$ and $1 \le a \le k$, the general formula

$$\operatorname{rank} B_{Q_a} = m - \gcd(2a, m)$$

specializes to $m - 2\gcd(a, m)$ because $m/\gcd(a, m)$ is even. The classifier reproduces the general formula in all tested cases. Representative rows are shown in Table 1.

field	a	Q	Arf	type	rank	formula
$\mathbb{F}_{2^4}$	1	$\operatorname{Tr}(x^3)$	1	O^-	2	2
$\mathbb{F}_{2^8}$	1	$\operatorname{Tr}(x^3)$	1	O^-	6	6
$\mathbb{F}_{2^8}$	2	$\operatorname{Tr}(x^5)$	1	O^-	4	4
$\mathbb{F}_{2^{16}}$	1	$\operatorname{Tr}(x^3)$	1	O^-	14	14
$\mathbb{F}_{2^{16}}$	4	$\operatorname{Tr}(x^{17})$	1	O^-	8	8
$\mathbb{F}_{2^{32}}$	1	$\operatorname{Tr}(x^3)$	1	O^-	30	30

Table 1: Representative output of `experiments/trace_form_arf.py`. The script checks all 15 cases with $m = 2, 4, 8, 16, 32$ and $1 \le a \le \log_2 m$.

Most of these forms have a nontrivial radical. The statement is therefore not "nondegenerate Gold form" but "positive-rank quadratic form with a nonsingular core whose Arf value is computed after symplectic reduction."

3.2 A broader trace family

The coefficient-1 Gold form is only one trace monomial. A standard trace description of the quadratic part of Boolean functions over $\mathbb{F}_{2^m}$ uses terms

$$\mathrm{Tr}(c_i x^{1+2^i})$$

plus affine terms and, in even degree, a middle half-trace term. The monomials displayed here are still game-operation-built in the nimber backend: $c_i x^{1+2^i} = c_i \otimes x \otimes x^{2^i}$, followed by trace/XOR.

The current implementation does not claim to package every Boolean quadratic function. The probe `experiments/gold_family_survey.py` deliberately omits the middle term and affine bookkeeping. What it does check, exhaustively over $\mathbb{F}_{256}$, is that scaled Gold components

$$Q_{\lambda,a}(x) = \mathrm{Tr}(\lambda x^{1+2^a})$$

already reach nondegenerate (bent) forms for many λ. For APN Gold exponents in that field, the observed count matches the classical $2(2^m-1)/3$ bent-component count. This matters for the open game question because a bent form has trivial radical: the loopy radical route collapses to $\{0\}$, and the frame-blind symplectic no-go applies without a degenerate layer to hide in.

Remark 2 (a vacuous small target)**.** At $m = 4$, $a = 1$ the zero set $\{Q_1 = 0\} \subset \mathbb{F}_{16}$ is the subfield $\mathbb{F}_4$, an $\mathbb{F}_2$-*subspace* of dimension 2: the form is genuinely quadratic (polar rank 2), but its zero set is affine, so as a target for "is this P-set a genuine quadric" it is vacuous – a linear rule realizes it. The smallest honest unscaled instance is $(8, 1)$. The repository's quadric-fitting layer (`src/forms/quadric_fit.rs`) reports exactly this distinction (genuine polar rank versus affine flat), and every probe below that consumes a fitted quadric is read with that guard in mind.

4 Arf as conditional win-bias

The relevant counting theorem is classical. If Q is nonsingular on $\mathbb{F}_2^{2r}$, then

$$\#\{x : Q(x) = 0\} = 2^{2r-1} + (-1)^{\mathrm{Arf}(Q)} 2^{r-1}.$$

For degenerate forms, the code uses the standard radical-adjusted count: an anisotropic radical balances the values, while an isotropic radical scales the bias of the nonsingular core.

The game interpretation is conditional:

> If a game has P-positions exactly $\{x : Q(x) = 0\}$, then the Arf invariant is the sign of the bias between second-player wins and first-player wins.

This is not a claim that such a game has been found. It is a target for a game semantics.

5 The linear ceiling

For a normal-play disjunctive sum of impartial games, the P-condition is linear in Grundy coordinates:

$$g_1 \oplus \cdots \oplus g_n = 0.$$

So normal-play sums give linear P-sets, not genuine quadrics. The coin-turning architecture – the very architecture whose product theorem defines the nim product – is subject to the same ceiling.

field	a	Arf	rank	$\#\{Q=0\}$	bias
$\mathbb{F}_{2^4}$	1	1	2	4	-4
$\mathbb{F}_{2^8}$	1	1	6	112	-16
$\mathbb{F}_{2^8}$	2	1	4	96	-32
$\mathbb{F}_{2^{16}}$	1	1	14	32512	-256
$\mathbb{F}_{2^{16}}$	4	1	8	30720	-2048

Table 2: Representative zero-counts from `experiments/arf_win_bias.py`. The bias is $\#\{Q = 0\} - 2^{m-1}$.

Theorem A (coin-turning linearity; standard math, machine cross-checked)**.** *Every Winning Ways coin-turning game has configuration P-set equal to the kernel of an $\mathbb{F}_2$-linear nimber-valued map. In particular no coin-turning P-set is a quadric of positive polar rank.*

Proof sketch. A coin-turning position is a finite set of heads; by the Winning Ways coin-turning theory [3], the Grundy value of a configuration is the XOR of the single-coin Grundy values $g(i)$ of its heads. The P-set is therefore $\{x : \bigoplus_{i\in x} g(i) = 0\}$, the kernel of the $\mathbb{F}_2$-linear map $x \mapsto \bigoplus_i x_i\, g(i)$. The repository cross-check (`experiments/gold/octal_attack.py`) verifies XOR-additivity by mex recursion, exhaustively over all 32,768 companion-family choices on 4 coins. □

Theorem A has a sharp classical complement: the linear ceiling is exactly where coding theory already harvests games.

Remark 3 (the lexicode shadow; standard math and interpretation)**.** Conway–Sloane lexicodes [5] are built by the greedy lexicographic rule – the mex rule – and the 1986 paper realizes them as the Grundy-value-0 positions of associated turning-game move structures; binary lexicodes are linear *because of* Sprague–Grundy theory (XOR-closure is a game theorem there, not a coding theorem). They include the Hamming codes and the length-24, $d = 8$ extended binary Golay code; over base 2^k they are closed under nim-addition, and they are linear at the Fermat-power bases 2^{2^k} – exactly the sizes at which nim-multiplication makes the ordinals below the base a field. Read together with Theorem A: lexicodes are the *floor* (rich linear codes naturally realized as Grundy-0 sets) and Theorem A is the *ceiling* (coin-turning realizes only linear P-sets); floor and ceiling meet at linear. The open problem of this note is whether the lexicode phenomenon admits a quadratic refinement, and the obstruction is precisely the XOR-closure failure that the polar form measures.

For a quadratic form in characteristic 2,

$$Q(u + v) = Q(u) + Q(v) + B(u, v).$$

Thus for $u, v \in \{Q = 0\}$,

$$u + v \in \{Q = 0\} \iff B(u, v) = 0.$$

The polar form is exactly the obstruction to XOR-closure. The first two layers are game-built: XOR is disjunctive sum and B is built from nim products. The missing layer is a play rule that integrates that bilinear data into the quadratic zero set.

6 Routes and obstructions

6.1 Test benches

The repo contains several probes. They are useful for narrowing the target, but they are not a proof that no natural game exists.

- `fit_f2_quadratic` tests whether a subset of $\mathbb{F}_2^k$ is the zero set of a quadratic polynomial, and whether the result has nonzero polar rank (distinguishing genuine quadrics from affine flats; cf. Remark 2).
- A bounded misere quotient computer tests small quotient P-sets. The octal sweep checked 292 octal codes, heap cutoff at most 4, and found quotient orders $2, 6, 10, 12, 14$, with no elementary abelian 2-group of rank at least 2. Theorem C below shows this emptiness is forced, not a sampling accident.
- `experiments/misere_kernel.py` checks the kernel obstruction on the smallest wild misere quotient R_8: the canonical group/kernel shadow is linear, while the genuinely misere P-element lies outside that vector-space part. Corollary 1 below shows the linearity is forced by commutativity.
- An interactive-kernel solver tests finite move graphs. It confirms that an ad hoc acyclic game can realize any subset, and that a rule referencing Q itself realizes $\{Q = 0\}$ tautologically. The tested B-coupled rules do not produce the Gold zero set.
- `examples/loopy_quadric.rs` tests cyclic move graphs with Draw sets. The symmetric B-coupled rule has Loss-set equal to the radical $R(B)$, so it reproduces the obstruction rather than the Gold zero set; this is the special case of Theorem G below, and its small $(4, 1)$ "hit" is the radical coincidence $|\{Q_1 = 0\}| = 4 = |R(B)|$ there.
- `examples/bent_route.rs` repeats route probes on a bent scaled Gold component. In that example, a B plus coordinate-frame rule reaches a quadric of the correct Arf class but not the specific Gold zero set; the naive local-field/spin-flip completion leaves the quadric variety.

Except for the frame-blind proposition below and the lettered theorems of Section 8, these are bounded or example-level negative results.

6.2 Frame-blind no-go

There is a clean obstruction for rules that see only a nondegenerate symplectic form and no frame.

Proposition 1. *Let B be a nondegenerate alternating form on $V = \mathbb{F}_2^{2r}$ with $r \geq 2$. If a finite normal-play game has position set V and a move relation invariant under $\mathrm{Sp}(B)$, then its P-set is not a nondegenerate quadric.*

Proof. Each element of $\mathrm{Sp}(B)$ is an automorphism of the move graph, so the retrograde outcome function is $\mathrm{Sp}(B)$-invariant. The P-set is therefore a union of symplectic orbits. For $r \geq 2$, the symplectic group is transitive on $V \setminus \{0\}$ [14], so the invariant subsets are only $\varnothing$, $\{0\}$, $V \setminus \{0\}$, and V. A nondegenerate quadric in dimension at least 4 has $2^{2r-1} \pm 2^{r-1}$ points, which is none of these. $\square$

The hypothesis matters. Coordinate-dependent rules have already broken $\mathrm{Sp}(B)$ symmetry, so this proposition does not explain every negative probe. It explains only genuinely frame-blind B-only rules. Section 8 extends the obstruction far beyond this symmetric class.

6.3 The diagonal framing

Once a coordinate frame is allowed, a canonical-looking quadratic refinement of B is

$$Q_{\mathrm{frame}}(v) = \sum_{i<j} B(e_i, e_j) v_i v_j.$$

For the Gold polar forms tested in this repo, the Gold form decomposes as

$$Q_a(v) = Q_{\mathrm{frame}}(v) + \ell_{\mathrm{diag}}(v), \qquad \ell_{\mathrm{diag}}(v) = \sum_i Q_a(e_i) v_i.$$

In the tested genuinely quadratic Gold cases up to $\mathbb{F}_{2^{16}}$, Q_{frame} has Arf 0 and the diagonal term changes the form to the Gold Arf-1 class.

Remark 4. This is an experimental pattern for the Gold polar forms tested here, not a theorem about every alternating form plus every coordinate frame. Random nondegenerate alternating forms can give a zero-diagonal frame quadric of Arf 1.

The diagonal $q_i = Q_a(e_i)$ is therefore the named missing datum. Two results of the parallel run locate it more precisely.

Proposition 2 (game-native diagonal source for $a = 1$; implemented and tested)**.** *Let $m = 2^k$ and let $w \in \mathbb{F}_{2^m}$ be the XOR of the Fermat coins 2^{2^t} for $t = 1, \ldots, k-1$. Then the $a = 1$ Gold diagonal satisfies*

$$q_i^{(m,1)} = Q_1(e_i) = \mathrm{Tr}\big(\wp(w)\, e_i\big), \qquad \wp(w) = w \otimes w \oplus w,$$

i.e. the trace-dual element of the diagonal is $\wp(w)$. Verified at $m = 4, 8, 16, 32$ via the trace-pairing uniqueness of the dual element (`experiments/gold/gold_diag_probe.py`).

This gives a game-native, tower-uniform, commutative source for the $a = 1$ diagonal: one nim-squaring, one XOR, one trace – all inside the licensed operation chain of Section 3. For even a the dual $\lambda_a^{(m)}$ provably drifts at each tower doubling (λ_2: $0, 6, 102, 24582$; λ_4: $31, 8030$), and no $\wp$-preimage family has been named beyond $m = 8$; the even-a analogue is open.

Remark 5 (subfield vanishing; standard math)**.** $q_i^{(m,a)} = 0$ for all $i < m/2$: the Fermat basis elements at lower indices, their Frobenius iterates, and their nim products stay inside the half-field, whose absolute trace from an even-degree extension vanishes. The Gold diagonal lives on the top Fermat layer only.

Open question. *Is the diagonal framing*

$$q_i = Q_a(e_i) = \mathrm{Tr}(e_i^{1+2^a})$$

itself game-natural for all a, in a way that yields a natural game with P-set $\{x : Q_a(x) = 0\}$? Proposition 2 answers the sourcing half affirmatively for $a = 1$; whether any acceptable source yields a non-tautological play rule is the subject of the remaining sections.

7 The extraspecial reframing

The diagonal-framing question can be restated in group-theoretic terms, and the restatement sharpens it. The dictionary is classical – the characteristic-2 Heisenberg/extraspecial picture of quadratic forms [13, 9] – and this section proves the pieces we use, to keep the note self-contained. Claim levels: Lemmas 1, 2, 3 and Proposition 3 are standard mathematics; the R_8 instantiation in Corollary 1 is implemented and tested (`experiments/misere_kernel.py`); reading E-equivariance as the right naturality criterion is interpretation; the existence of a game-native model of the extension is open.

Throughout, $V = \mathbb{F}_2^n$ written additively, and $Z = \{1, z\} \cong \mathbb{Z}/2$ written multiplicatively. A *quadratic map* $Q\colon V \to \mathbb{F}_2$ is a function with $Q(0) = 0$ whose polarization $B(x, y) = Q(x + y) + Q(x) + Q(y)$ is bilinear; B is then alternating (hence symmetric), and Q may have any diagonal $q_i = Q(e_i)$, exactly the char-2 datum the engine keeps separate from b.

7.1 Quadratic forms are central extensions

Lemma 1 (extraspecial dictionary). *Let $1 \to Z \to E \xrightarrow{\pi} V \to 1$ be a central extension of the elementary abelian 2-group V by $Z \cong \mathbb{Z}/2$.*

(i) For $x \in V$ and any lift $\tilde{x} \in \pi^{-1}(x)$ one has $\tilde{x}^2 \in Z$, and $\tilde{x}^2$ is independent of the lift; the squaring form $Q_E\colon V \to \mathbb{F}_2$ defined by $\tilde{x}^2 = z^{Q_E(x)}$ is well defined, as is the commutator pairing B_E defined by $[\tilde{x}, \tilde{y}] = z^{B_E(x,y)}$, which is bilinear and alternating. They satisfy

$$(\tilde{x}\tilde{y})^2 = \tilde{x}^2\, \tilde{y}^2\, [\tilde{x}, \tilde{y}], \qquad \textit{equivalently} \qquad Q_E(x + y) = Q_E(x) + Q_E(y) + B_E(x, y),$$

so Q_E is a quadratic map with polar form B_E.

(ii) Conversely, every quadratic map Q arises: fixing a basis $e_1, \ldots, e_n$ of V, the bilinear 2-cocycle

$$\varphi(x, y) = \sum_i x_i y_i\, Q(e_i) + \sum_{i>j} x_i y_j\, B(e_i, e_j)$$

defines a group $E_Q = Z \times V$ with $(s, x)(t, y) = (s + t + \varphi(x, y),\ x + y)$, whose squaring form is Q and whose commutator pairing is the polar form B.

(iii) The assignment $Q \mapsto [E_Q]$ is a bijection from quadratic maps on V to equivalence classes of central extensions of V by $\mathbb{Z}/2$.

(iv) E_Q is abelian iff $B = 0$ (iff Q is linear); and for $n \geq 1$, E_Q is extraspecial (i.e. $Z(E) = [E, E] = \Phi(E) \cong \mathbb{Z}/2$) iff B is nondegenerate, which forces $n = 2r$ even and $|E_Q| = 2^{1+2r}$.

Proof. (i) $\pi(\tilde{x}^2) = 2x = 0$, so $\tilde{x}^2 \in Z$; replacing $\tilde{x}$ by $z\tilde{x}$ multiplies the square by $z^2 = 1$. Commutators of lifts lie in Z because V is abelian, are central, and are unchanged by central retagging of the lifts; centrality of commutators gives bimultiplicativity, and $[\tilde{x}, \tilde{x}] = 1$ gives $B_E(x, x) = 0$. For the identity: with $c = [\tilde{y}, \tilde{x}]$ central, $\tilde{x}\tilde{y}\tilde{x}\tilde{y} = \tilde{x}(\tilde{x}\tilde{y}\, c)\tilde{y} = \tilde{x}^2\tilde{y}^2 c$, and $c = c^{-1} = [\tilde{x}, \tilde{y}]$ since Z has exponent 2. As $\tilde{x}\tilde{y}$ lifts $x + y$, the additive identity follows.

(ii) A bilinear map is a 2-cocycle, so E_Q is a group with center containing $Z = \{(0, 0), (1, 0)\}$. Squares: $(s, x)^2 = (\varphi(x, x), 0)$ and $\varphi(x, x) = \sum_i x_i Q(e_i) + \sum_{i>j} x_i x_j B(e_i, e_j) = Q(x)$ by the polarization expansion of Q in the basis. Commutators: $(s, x)(t, y)(s, x)^{-1}(t, y)^{-1} = (\varphi(x, y) + \varphi(y, x), 0) = (B(x, y), 0)$, using that the symmetrization of φ is B (the diagonal terms cancel and B is symmetric).

(iii) Equivalent extensions have cohomologous cocycles, and a coboundary $\delta\psi(x,y) = \psi(x) + \psi(y) + \psi(x+y)$ has zero diagonal ($\delta\psi(x,x) = \psi(0) = 0$), so the squaring form is an invariant of the class; (ii) gives surjectivity. Injectivity: if two extensions have equal squaring form Q, choose sections and cocycles φ, φ'; both have diagonal Q, and both have symmetrization equal to the polar form of Q, so $d = \varphi + \varphi'$ is a symmetric cocycle with zero diagonal. The extension E_d is then abelian of exponent 2, i.e. an $\mathbb{F}_2$-vector space, so $1 \to Z \to E_d \to V \to 1$ splits and d is a coboundary; the two extensions are equivalent.

(iv) Commutators generate $z^{\operatorname{im} B}$, so E_Q is abelian iff $B = 0$. In general $Z(E_Q) = \pi^{-1}(\operatorname{rad} B)$, since $\tilde{x}$ is central iff $B(x, \cdot) = 0$; thus $Z(E_Q) = Z$ iff B is nondegenerate. If so, $B \neq 0$ gives $[E,E] = Z$, and $\Phi(E) = E^2[E,E] = Z$ because all squares lie in Z and E/Z is elementary abelian. A nondegenerate alternating form exists only in even dimension. $\square$

Remark 6. This is the characteristic-2 Heisenberg picture: E_Q is the Heisenberg group of (V, B) and Q selects the central extension; cohomologically, (iii) is the standard identification of $H^2(V; \mathbb{F}_2)$ with quadratic maps $V \to \mathbb{F}_2$ [13]. The same finite-quadratic-module data drives the Weil S/T matrices in the library's integral layer. Note that the dictionary is exactly the engine's discipline in group clothing: the diagonal q_i (squares) and the off-diagonal b_{ij} (commutators) are independent central data, and collapsing them is collapsing E_Q onto an abelian quotient.

7.2 Arf classifies the two extraspecial types

Write H for the hyperbolic plane ($Q(e_1) = Q(e_2) = 0$, $B(e_1, e_2) = 1$; $\mathrm{Arf} = 0$) and A for the anisotropic plane ($Q \equiv 1$ on $V \setminus \{0\}$; $\mathrm{Arf} = 1$), and $\circ$ for the central product (identify the centers).

Lemma 2 (Arf = the two types)**.** *Let B be nondegenerate,* $\dim V = 2r \geq 2$.

(i) $E_{Q \perp Q'} \cong E_Q \circ E_{Q'}$.

(ii) $E_H \cong D_8$ *and* $E_A \cong Q_8$.

(iii) Every extraspecial 2*-group of order* 2^{1+2r} *is isomorphic to* E_Q *for a nondegenerate* Q*, and* $E_Q \cong E_{Q'}$ *iff* $(V,Q) \cong (V',Q')$ *iff the ranks and Arf invariants agree [6, 1]. Hence there are exactly two extraspecial groups of order* 2^{1+2r}*:*

$$E^+_{2^{1+2r}} = D_8^{\circ r} \ (\mathrm{Arf} = 0,\ Q \cong H^{\perp r}), \qquad E^-_{2^{1+2r}} = D_8^{\circ(r-1)} \circ Q_8 \ (\mathrm{Arf} = 1,\ Q \cong H^{\perp(r-1)} \perp A).$$

In particular $D_8 \circ D_8 \cong Q_8 \circ Q_8$*, the group avatar of* $H \perp H \cong A \perp A$ *(additivity of* Arf*).*

(iv) (Census.) With $\varepsilon = (-1)^{\mathrm{Arf}\, Q}$,

$$\#\{g \in E_Q : g^2 = 1\} = 2\,\#\{x : Q(x) = 0\} = 2^{2r} + \varepsilon\, 2^r,$$

so the element-order census of E_Q *is the zero-count bias of Section 4 read multiplicatively, and it distinguishes the two types.*

Proof. (i) The cocycle $\varphi \oplus \varphi'$ on $V \oplus V'$ is bilinear with diagonal $Q \perp Q'$, and $(s, (x, x')) \mapsto [((s,x), (0,x'))]$ is an isomorphism onto $(E_Q \times E_{Q'})/\langle (z, z') \rangle$.

(ii) E_H has order 8, is nonabelian ($B \neq 0$), and has a noncentral involution (any lift of e_1 squares to $z^{Q(e_1)} = 1$), so $E_H \cong D_8$; in E_A every noncentral element squares to z (as $Q \equiv 1$ off 0), so z is the unique involution and $E_A \cong Q_8$.

(iii) For E extraspecial, $V = E/Z(E)$ is elementary abelian (as $\Phi(E) = Z(E)$), so Lemma 1 applies to $1 \to Z(E) \to E \to V \to 1$ and gives $E \cong E_{Q_E}$ with B_{Q_E} nondegenerate ($Z(E)$

exactly central). If $\psi\colon E_Q \to E_{Q'}$ is an isomorphism, it preserves the (characteristic) centers, fixes $z \mapsto z'$ (the unique nontrivial central element), and induces an $\mathbb{F}_2$-linear map $\bar{\psi}\colon V \to V'$ with $Q'(\bar{\psi}x)$ read from $\psi(\tilde{x})^2 = \psi(\tilde{x}^2)$ – an isometry. Conversely an isometry g transports the cocycle: $\varphi' \circ (g \times g)$ has diagonal $Q' \circ g = Q$, hence by Lemma 1(iii) yields an extension equivalent to E_Q, and $(s, x) \mapsto (s, gx)$ closes the isomorphism. Dickson's classification of nonsingular quadratic forms over $\mathbb{F}_2$ (two isometry classes per even rank, separated by Arf) finishes; the central-product normal forms follow from (i), (ii) and Witt decomposition.

(iv) The two lifts of x both square to $z^{Q(x)}$, so they are involutions or the identity iff $Q(x) = 0$; now apply the zero-count formula of Section 4. The counts differ for $\varepsilon = \pm 1$, so $E^+ \ncong E^-$. □

7.3 The abelian obstruction

Lemma 3 (abelian obstruction)**.** *Let M be a commutative monoid, $Z = \{1, z\} \subseteq M$ a two-element subgroup, $N \subseteq M$ a submonoid containing Z, and $\pi\colon N \twoheadrightarrow V$ a surjective monoid homomorphism onto an $\mathbb{F}_2$-vector space with $\pi^{-1}(\pi(m)) = mZ$ for all $m \in N$. Then the squaring form $Q\colon V \to \mathbb{F}_2$, defined by $\tilde{m}^2 = z^{Q(x)}$ for any $\tilde{m} \in \pi^{-1}(x)$, is well defined and $\mathbb{F}_2$-linear; its polar form vanishes identically. Consequently no commutative monoid realizes, through its own squaring map, a quadratic form with nonzero – a fortiori nondegenerate – characteristic-2 polar form. Equivalently, by Lemma 1(iv): if $B \neq 0$, the extension E_Q is nonabelian and admits no model (N, Z, π) inside any commutative monoid.*

Proof. $\pi(m^2) = 2\pi(m) = 0$, so $m^2 \in \pi^{-1}(0) = Z$, and $(zm)^2 = z^2 m^2 = m^2$, so Q is well defined. Commutativity gives $(mn)^2 = m^2 n^2$; since $\tilde{m}\tilde{n}$ is a preimage of $x+y$, this is $Q(x+y) = Q(x)+Q(y)$, i.e. $B \equiv 0$. □

Corollary 1 (misère quotients: the kernel shadow is linear)**.** *Let $\mathcal{Q}$ be a finite misère quotient – a finite commutative monoid – with kernel K, the maximal subgroup of $\mathcal{Q}$ (the mutual-divisibility class of the product of all idempotents) [12].*

- *(i) Every configuration (N, Z, π) as in Lemma 3 inside $\mathcal{Q}$ has linear squaring form: the intrinsic multiplication of a misère quotient cannot supply a quadratic refinement with $B \neq 0$, on K or anywhere else in $\mathcal{Q}$.*
- *(ii) Under the regularity hypothesis of [12, Thm. 6.4] (satisfied by the regular finite quotients arising in practice), $K \cong (\mathbb{Z}/2)^k$ is the group of normal-play Grundy values and the P-portion meets K in the XOR-linear normal-play set.*
- *(iii) On the smallest wild quotient $R_8 = \langle a, b, c \mid a^2 = 1,\ b^3 = b,\ bc = ab,\ c^2 = b^2 \rangle$ with $P = \{a, b^2\}$: the idempotents are $\{1, b^2\}$, the kernel is $K = \{b^2, b, ab, ab^2\} \cong (\mathbb{Z}/2)^2$ with identity b^2, $P \cap K = \{b^2\} \mapsto \{0\}$ is linear, and the genuinely misère P-element a lies outside K. (Implemented and tested: `experiments/misere_kernel.py`.)*

So the linear obstruction observed on R_8 is forced by commutativity, not an accident of the example: a nondegenerate polar form is the commutator pairing of a nonabelian group, and a misère quotient has none to offer.

Proof. (i) is Lemma 3 applied to submonoids of $\mathcal{Q}$, which are commutative. (ii) is quoted from [12]. (iii) is a finite verification. □

7.4 The integral shadow

The same extension data has an integral-lattice avatar, already shipped in the library (interpretation, verified through the `DiscriminantForm` + `arf_nimber` pipeline): the Gold nonsingular core of rank $2r$ and Arf ε is the discriminant form of the even lattice $U(2)^r$ ($\varepsilon = 0$) or $U(2)^{r-1} \oplus D_4$ ($\varepsilon = 1$); the Milgram phase is $4 \cdot \mathrm{Arf} \mod 8$; the Weil S-matrix prefactor is $(-1)^{\mathrm{Arf}}$; and the Frobenius–Schur indicator of the Heisenberg representation of E_Q is $(-1)^{\mathrm{Arf}}$. The metaplectic relation $(ST)^3 = S^2$ holds iff the diagonal T uses a quadratic refinement of the correct Arf class – it selects the Arf *class*, not the member. So the integral bridge carries exactly one game-relevant bit; it cannot, by itself, distinguish the Gold form from any other refinement in its class.

7.5 A Tier-2 naturality screen: E-equivariance

Proposition 1 kills rules that are blind to everything but the symplectic structure (Tier 1), while per-position evaluator circuits for Q_a realize the quadric tautologically (Tier 3). The extraspecial dictionary suggests a criterion strictly between the two: the rule should see the extension E_Q – which carries the diagonal data q_i structurally, as squares – but only up to its automorphisms, so that no basis, field structure, or evaluation circuit can be smuggled in.

Definition 1 (uniform rules; the three tiers)**.** Let Q be nondegenerate on $V = \mathbb{F}_2^{2r}$ with polar form B, and $E = E_Q$, $\pi\colon E \to V$, $\Sigma = \{x \in V : Q(x) = 0\}$.

(a) A *uniform rule on a finite set* X is a single move relation $M \subseteq X \times X$, fixed once for all positions; for normal play we require the move digraph acyclic, and outcomes are computed by the usual retrograde recursion. (For loopy or misère readings, replace the outcome recursion; the equivariance constraint below applies verbatim, since outcome classes are invariants of digraph automorphisms.)

(b) A uniform rule on E is *E-equivariant* if every $\alpha \in \mathrm{Aut}(E)$ is an automorphism of the move digraph. Its *shadow* is $\pi(P) \subseteq V$, where P is its P-set; it *realizes* $T \subseteq V$ if $\pi(P) = T$.

(c) *Tier 1* ($\mathrm{Sp}(B)$-blind): a uniform rule on V invariant under $\mathrm{Sp}(B)$, as in Proposition 1. *Tier 3* (Q_a-evaluation): a family of games $\{\Gamma_x\}_{x\in V}$ whose structure varies with the designated input x (e.g. the trace-circuit evaluator); this is not a uniform rule. *Tier 2 screen*: a route to the Gold quadric is Tier-2 natural *only if* its positions and moves lift to an E-equivariant uniform rule on E_{Q_a} – with Q_a restricted to its nonsingular core when the form is degenerate, matching the classifier's radical discipline – that realizes $\{Q_a = 0\}$.

Proposition 3 (the screen sits strictly between the solved tiers)**.** *Let Q be nondegenerate on $V = \mathbb{F}_2^{2r}$, $r \geq 2$, and $E = E_Q$.*

(i) The image of $\mathrm{Aut}(E) \to \mathrm{GL}(V)$ *is exactly the orthogonal group* $\mathrm{O}(Q)$*, with kernel the central twists $g \mapsto g\, z^{\ell(\pi g)}$, $\ell \in V^*$, which equal* $\mathrm{Inn}(E) \cong V$ *because B is nondegenerate; thus* $\mathrm{Out}(E) \cong \mathrm{O}(Q)$ *(cf. [15]). The* $\mathrm{Aut}(E)$*-orbits on E are*

$$\{1\}, \qquad \{z\}, \qquad \pi^{-1}(\Sigma \setminus \{0\}), \qquad \pi^{-1}(V \setminus \Sigma).$$

(ii) Hence the P-set of any E-equivariant uniform rule is a union of these four orbits, and its shadow is one of the eight unions of $\{0\}$, $\Sigma \setminus \{0\}$, $V \setminus \Sigma$. The quadric Σ is among them: the Tier-1 exclusion does not apply.

(iii) (Tier 1 $\subsetneq$ Tier 2.) Every $\mathrm{Sp}(B)$*-invariant uniform rule on* V *pulls back along* π *to an* E*-equivariant rule with* P*-set* π^{-1} *of the original, whose shadow is therefore one of* $\varnothing$, $\{0\}$, $V \setminus \{0\}$, V *– never the quadric (Proposition 1 in this language). The containment is strict: the squaring rule*

$$g \to h \text{ legal} \iff g^2 = z \text{ and } h^2 = 1$$

("move from any order-4 position to any position of order at most 2") is E*-equivariant and acyclic, and its* P*-set is the involution locus* $\{g : g^2 = 1\} = \pi^{-1}(\Sigma)$*, with shadow exactly* Σ*.*

(iv) (Tier 2 $\subsetneq$ Tier 3.) Up to the isometry induced by any group isomorphism, the family of E*-equivariant rules and their shadows depends only on* $(r, \mathrm{Arf}\, Q)$ *(Lemma 2(iii)). In particular the screen cannot separate Gold forms of equal core rank and Arf invariant, and an* E*-equivariant rule has no access to* m*, the exponent* a*, the field multiplication, the coordinate frame* $q_i = Q_a(e_i)$*, or any evaluation circuit. Conversely, arbitrary subsets of* V *– all realizable by ad hoc acyclic lookup games and by Tier-3 evaluators (Section 6.1) – are not shadows of* E*-equivariant rules once they leave the eight-set list of (ii).*

Proof. (i) Automorphisms preserve $Z = Z(E)$ and fix z ($\mathrm{Aut}(\mathbb{Z}/2) = 1$), so the induced map preserves the squaring form: $Q(\bar{\alpha}x)$ is read from $\alpha(\tilde{x})^2 = \alpha(\tilde{x}^2) = z^{Q(x)}$; the image lies in $\mathrm{O}(Q)$. Surjectivity: an isometry g transports the standard cocycle as in the proof of Lemma 2(iii), producing an automorphism of E_Q inducing g. The kernel consists of maps $g \mapsto g\, z^{\ell(\pi g)}$ with ℓ linear (multiplicativity forces additivity of ℓ); inner automorphisms are exactly the twists $\ell = B(\bar{g}, \cdot)$, and nondegeneracy makes $\bar{g} \mapsto B(\bar{g}, \cdot)$ onto V^*. Orbits: $\mathrm{O}(Q)$ is transitive on $\Sigma \setminus \{0\}$ and on $V \setminus \Sigma$ by Witt's extension theorem in its characteristic-2 quadratic-space form [14] (both sets are nonempty for $r \geq 2$); the two lifts of any $x \neq 0$ are conjugate under conjugation by $\tilde{g}$ with $B(\bar{g}, x) = 1$; and 1, z are fixed by everything.

(ii) Digraph automorphisms preserve the retrograde outcome recursion, so P is $\mathrm{Aut}(E)$-invariant; apply (i) and project.

(iii) The pullback $M = \{(g, h) : (\pi g, \pi h) \in R\}$ of an $\mathrm{Sp}(B)$-invariant rule R is preserved by $\mathrm{Aut}(E)$, since the induced action on V lies in $\mathrm{O}(Q) \subseteq \mathrm{Sp}(B)$; it is acyclic when R is, and an easy induction along the acyclic rank gives $P(M) = \pi^{-1}(P(R))$. By the transitivity of $\mathrm{Sp}(B)$ on $V \setminus \{0\}$ for $r \geq 2$, $P(R)$ is a union of $\{0\}$ and $V \setminus \{0\}$. For the squaring rule: automorphisms preserve element orders, so the rule is E-equivariant; positions with $g^2 = 1$ are terminal, hence P; positions with $g^2 = z$ have a move (to 1), hence N. Its shadow Σ satisfies $1 < |\Sigma| = 2^{2r-1} + \varepsilon 2^{r-1} < 2^{2r} - 1$ for $r \geq 2$, so it is none of the four Tier-1 shadows.

(iv) If $\psi \colon E_Q \to E_{Q'}$ is an isomorphism (it exists iff ranks and Arf agree, Lemma 2(iii)), then $M \mapsto (\psi \times \psi)(M)$ is a bijection between E-equivariant rules carrying P-sets to P-sets and shadows to shadows through the induced isometry $\bar{\psi}$; in particular it carries rules realizing $\{Q = 0\}$ to rules realizing $\{Q' = 0\}$. The shadow constraint of (ii) excludes all but eight subsets, while Tier-3 constructions realize any subset; the containment is proper. $\square$

Remark 7 (what the screen does and does not settle)**.** Equivariance is a screen, not a construction. Over the *abstract* group E the screen is satisfiable – Proposition 3(iii) exhibits the squaring rule, which is nothing but the extraspecial squaring map of Lemma 1 read as a one-move relation. That rule consults only the multiplication of E, so it is a Q-evaluator exactly insofar as E itself is fed in by hand. The reframing therefore relocates the open problem: instead of *find a play rule that computes* Q_a, it asks *exhibit a game-native model of the extension* E_{Q_a} – positions built from game values, multiplication from game constructions – after which the rule layer is canonical. Lemma 3 is the sharp constraint on that search: no commutative value-world, in particular no misère-quotient kernel (Corollary 1), can host E once $B \neq 0$. The noncommutativity must enter from a structurally available asymmetry, and the one that normal, misère, and partizan play all possess – and that the symmetric polar form B discards – is the first-/second-player asymmetry of the move relation. Claim levels: the proposition is standard mathematics; treating E-equivariance as *the* naturality criterion is interpretation; the existence of a game-native E is open. (For $r = 1$ the screen degenerates: on the anisotropic plane $\Sigma = \{0\}$ and $\mathrm{O}(Q) = \mathrm{Sp}(B)$, so the statement is vacuous there; the Gold targets of interest have $r \geq 2$ cores.) Theorem E below sharpens the screen further: read as *invariance* it admits no live middle at all, so the E-structure must enter *covariantly*, in the dynamics.

Open question. *Does any game-native construction realize the extraspecial extension* E_{Q_a} *– equivalently, produce the diagonal data* $q_i = Q_a(e_i)$ *as squares in a noncommutative game-built structure – without an evaluation circuit for* Q_a*? By Lemma 3 the source cannot be any commutative game-value monoid.*

8 No-go theorems for the middle tier

This section records the no-go results of the structured attack (2026-06-10) on the middle tier. Architectures covered: commutative provenance, affine and orthogonal symmetry, bounded B-oracle access, B-local flip rules, and E-translation dynamics. Verification scopes are stated inline; the supporting probes live in `experiments/gold/`. (Theorem A, coin-turning linearity, opened the series in Section 5.)

Theorem B (commutative squaring collapse; standard math)**.** *In any commutative monoid, squaring is an endomorphism, so any quadratic datum read through homomorphic coordinates has polarization identically zero. In particular the extraspecial squaring map* Q *(for* $B \neq 0$*) cannot live in, map into, or pull back through any commutative game monoid. This covers misère quotients, turn-parity counters, and all coin-turning configuration sums.*

Proof. Immediate from Lemma 3: commutativity gives $(mn)^2 = m^2 n^2$, so the squaring form is additive and $B \equiv 0$. $\square$

Theorem C (group misère quotients are tiny; conditional)**.** *Assume [12, Thm. 6.4] with its regularity hypothesis. If the misère indistinguishability quotient* $\mathcal{Q}(\mathcal{A})$ *of a set of impartial games is a group, then* $|\mathcal{Q}(\mathcal{A})| \leq 2$*.*

Proof sketch. Doubling lemma: in a group quotient every element is cancellable, and any position of Grundy value 2 doubles to a misère-P position, contradicting Grundy-distinctness from 0; the quoted structure theorem then pins the quotient to at most $\{1, a\}$. $\square$

Remark 8 (scope and a probe caveat)**.** Theorem C is conditional on the Plambeck–Siegel structure theorem as cited; the regularity hypothesis is quoted from the repository's `experiments/misere_kernel.py` docstring, and the JCTA 2008 paper has *not* been independently re-verified within this program (closing that gate is a listed next move). Two consequences for the probe map: the $(\mathbb{Z}/2)^k$ ($k \geq 2$) target of `examples/octal_hunt.rs` is empty a priori – its empirical negatives (quotient orders $2, 6, 10, 12, 14$) are a theorem, not a sampling accident – and any future "hit" from that sweep would be a labeling artifact, since the helper `p_set_as_f2` checks only that the subset labeling is a bijection, not that it is a monoid homomorphism. The hunt's target should be reframed accordingly.

For the remaining results, fix a nondegenerate quadratic form Q on $V = \mathbb{F}_2^{2r}$, $r \geq 2$, with polar form B and zero set $\Sigma = \{Q = 0\}$, and write $\mathrm{AO}(Q) = \{x \mapsto gx + c : g \in \mathrm{Sp}(B),\ Q(gx + c) = Q(x)\ \forall x\}$ for the affine stabilizer appearing below.

Theorem D (affine symmetry budget)**.** $\mathrm{Stab}_{\mathrm{AGL}(V)}(\Sigma) = \mathrm{AO}(Q)$*; it contains no nontrivial pure translation (for a nonsingular core,* $c \in \mathrm{rad}(B) = 0$*), and its pure-linear part is exactly* $\mathrm{O}(Q)$*. Consequently every uniform rule whose move digraph admits even one affine automorphism outside* $\mathrm{AO}(Q)$ *has P-set $\neq \Sigma$.*

Verification. Exhaustive over all 322,560 affine maps of $\mathbb{F}_2^4$ at $m = 4$, both Arf classes (`experiments/gold/skeptic_nogo_check.py`); the linear-part identification is standard math (Witt extension).

Theorem E (the symmetry dial has no middle)**.** $\mathrm{O}(Q)$ *is maximal in* $\mathrm{Sp}(B)$*, and the* $\mathrm{O}(Q)$*-orbitals on $V \times V$ are exactly the Gram classes $(Q(u), Q(w), B(u, w))$ together with the degeneracy flags (Witt extension; see [7, §8]). Hence a rule invariant under any group strictly larger than* $\mathrm{O}(Q)$ *is dead by Theorem D (its symmetry reaches outside* $\mathrm{AO}(Q)$*), while a rule fully* $\mathrm{O}(Q)$*- or* $\mathrm{Aut}(E)$*-equivariant factors extensionally through pointwise Q- and B-evaluation. There is no invariance class strictly between dead (too coarse) and Gram-factoring (an evaluator): E-naturality must be covariance, not invariance.*

Verification. Maximality verified at $m = 4$: every one of the 648 (resp. 600) elements of $\mathrm{Sp}(4, 2)$ outside $\mathrm{O}^+(4, 2)$ (resp. $\mathrm{O}^-(4, 2)$) generates $\mathrm{Sp}(4, 2)$ together with the orthogonal group; orbitals enumerated exhaustively at $m = 4$ (`experiments/gold/ao_orbitals.py`).

Theorem F (B-oracle lower bound)**.** *Access model: legality of a move is decided from XOR combinations of the two positions and t fixed constant vectors, with oracle access to B only. If $t \leq 2r-3$, then for every quadratic refinement Q' of B simultaneously there is a nonzero Q'-singular vector in the complement of the constant span; the transvection it induces is a digraph automorphism lying outside* $\mathrm{O}(Q')$*, and Theorem D kills the P-set. The bound is tight: at $t = 2r-2$ with anisotropic complement the argument fails. Hence the Gold diagonal framing $q_i = Q_a(e_i)$, which supplies $2r$ frame constants, is within two dimensions of information-theoretically forced.*

Verification. Transvection step and the exact escape boundary $t = 2r - 2$ machine-checked at $m = 4$; end-to-end spot check on 50 random $t = 1$ B-oracle rules, retrograde-solved (`experiments/gold/skeptic_nogo_check.py`, `nogo_verify.py`).

Theorem G (B-local flip rules)**.** *Any rule of the form "flip d at v iff $f(d, B(v, d))$" is automatically undirected, because B is alternating: $f(d, B(v + d, d)) = f(d, B(v, d) + B(d, d)) = f(d, B(v, d))$. For undirected loopy games the outcome partition collapses:* $\mathrm{Win} = \varnothing$, $\mathrm{Loss} =$ *the isolated vertices $=$ an affine flat, and* $\mathrm{Loss} \cup \mathrm{Draw} \neq \Sigma$ *for $r \geq 2$ (the count $|\Sigma| = 2^{r-1}(2^r + \varepsilon)$ has an odd factor > 1). This covers the Loss, Draw, and* $\mathrm{Loss} \cup \mathrm{Draw}$ *targets at once. The symmetric-B rule of* `examples/loopy_quadric.rs` *is the special case $f(d, \beta) = \beta$; its $(4, 1)$ "hit" is the radical coincidence of Section 6.1.*

Verification. Undirectedness is the two-line identity above (standard math); the loopy outcome collapse including the Draw-set gap is machine-checked at $m = 4$ (`experiments/gold/skeptic_nogo_check.py`).

Theorem H (E-translation no-go)**.** *For $r \geq 2$, no left-E-equivariant rule on $E = E_Q$ with moves $g \to gn$, $n \in N$ for a fixed nonempty $N \subseteq E$, has P-set equal to the involution locus $I = \{g : g^2 = 1\} = \pi^{-1}(\Sigma)$.*

Proof. $I \cdot I = E$: for every $v \neq 0$, the count $\#\{x : Q(x) = 0 = Q(x+v)\} = 2^{2r-2} + (-1)^{\operatorname{Arf} Q} 2^{r-1}$ is positive for $r \geq 2$ (character sum), so every $n \in N$ factors as $n = gh$ with $g, h \in I$. Then $g \in I$, and $g \cdot n = g(gh) = g^2 h = h \in I$: the move $g \to gn$ is an edge from I to I. A normal-play P-set admits no P-to-P edge, so $P \neq I$. □

Verification. Cross-checked on the bent $(8, 1, \lambda = 2)$ extraspecial group (`experiments/gold/extraspecial_core.py`).

8.1 Normal-play rigidity and the synthesis

The strongest constraint is not any single architecture kill but a rigidity statement inside a declared access class. Call a rule *in-class* if the quadratic refinement is quarantined behind a weight-bounded oracle (the (w_0, c) metering of Definition 2 below) and the rule is *refinement-uniform*: it realizes the target for every refinement q of B in the declared family, not just the Gold one.

Theorem 1 (normal-play rigidity, in-class)**.** *Under the bounded-framing oracle model with refinement uniformity, every legal edge between positions of Hamming weight $> c\,w_0$ must flip Q – in normal, misère, and loopy-Loss semantics alike. Consequently every in-class realizer of Σ in those semantics is a Q-alternating ender: play telescopes Q, and the outcome is the parity of $Q(x)$.*

Proof sketch. The case $Q(v) = Q(w) = 0$ of a non-flipping edge is forbidden directly by the outcome structure in all three readings. The case $Q(v) = Q(w) = 1$ is reduced to it by an adversary substitution: replace q by $q' = q + \ell$ with ℓ a linear form annihilating every queried functional but not the dense indicators; refinement uniformity transports the rule to q', where the edge becomes a forbidden 0-to-0 edge. □

Theorem 2 (no live middle, relative to the quarantine)**.** *Let Q have nonsingular core of rank $2r \geq 4$. No fixed uniform single-board rule satisfying*

- *(S) no affine digraph automorphism outside* $\mathrm{AO}(Q)$,
- *(A) legality from B-oracle queries at $t \leq 2r - 3$ frame constants,*
- *(C) no commutative squaring route for the provenance of Q,*
- *(E) no left-E-equivariant structure on the extraspecial group,*
- *(R) bounded-framing oracle access with refinement uniformity,*

has P/Loss/Draw target equal to Σ with any bulk strategic content: under (R), every legal edge between dense positions flips Q, all strategic content is confined to a Hamming ball of radius $c\,w_0$, and no bulk-decision-live realizer exists.

Remark 9 (honest scope: five escape hatches)**.** Theorem 2 is relative to its hypotheses, and each hypothesis is an open experimental window:

1. *Loopy-Draw semantics*: the substitution contradiction of Theorem 1 does not fire (Draw-to-Draw edges are legal).
2. *$t \geq 2r - 2$ with anisotropic complement*: escapes Theorem F at the symmetry level.
3. *Frobenius-aware access*: the Galois group lies inside $\mathrm{O}(Q_a)$, so all symmetry methods are silent; Theorem F's model explicitly excludes Frobenius-twisted queries.
4. *Non-quarantined rules using the game-native $\wp(w)$ diagonal source* (Proposition 2): once q is game-built, refinement uniformity holds trivially and class (R) says nothing.

5. *Rank $r = 1$ and radical-anisotropic degenerate layers*: the character-sum count in Theorem H vanishes, and small extraspecial groups can be realized.

The quarantine hypothesis (R) is a choice with a justification: without quarantining the form-identifying datum, the descent evaluator is decision-live everywhere and no purely complexity-based no-go is possible. Whether game-built diagonal sources such as $\wp(w)$ count as "probing" or "feeding in" the form is the central open definitional question of this program.

9 A formalized naturality criterion

Two natural first attempts at "naturality" fail for structural reasons. *Equivariance fails*: by Theorem E there is no invariance class strictly between symmetry groups that kill the P-set and symmetry groups that force extensional factoring through Q- and B-evaluation. *Vocabulary quarantine fails*: the Gold diagonal $q_i^{(m,1)} = \mathrm{Tr}(e_i^3)$ is expressible as $\mathrm{Tr}(\wp(w)e_i)$ using only XOR, nim-product, and trace (Proposition 2) – the licensed operation chain itself – so no principled line separates "computing q" from "computing Q" on vocabulary alone. The criterion below replaces both with access metering plus a strategic-content axiom.

Definition 2 (uniform local rules; the N-axioms)**.** Fix the coin frame $\{e_i\}$ (the Grundy basis $g(n) = 2^n$, game-natural). *Public data*: the frame, $\oplus$, nim-product, Frobenius, and the full alternating form B (game-built via Turning Corners, Frobenius, and trace). *Quarantined data*: the quadratic refinement, presented as a weight-bounded oracle $O_q(z) = Q_q(z)$ for $\mathrm{wt}(z) \le w_0$. A *uniform local rule* satisfies:

- **F1** (*q-blind statics*) position set, loading map, and terminal set are computed without oracle access;
- **F2** (*metered access*) each legality bit is decided with at most c adaptive oracle queries, each at a point of Hamming weight at most w_0, with (w_0, c) constants independent of m;
- **F3** (*declared semantics*) one canonical outcome labeling (normal, misère, loopy, or q-blind terminal payoff).

It is a *natural realizer* if additionally:

- **N1** (*torsor uniformity – anti-evaluator*) for every m, every B in the declared family, and every refinement q of B: $\mathrm{outcome}(\iota(x)) = \mathrm{target}$ iff $Q_q(x) = 0$;
- **N2** (*locality budget*) the (w_0, c) metering of F2;
- **N3** (*strategic relevance – anti-clock*) for every instance (m, B, q) there is a reachable position that is simultaneously *outcome-critical* (it has legal moves to different outcome classes) and *form-live* (its outcome differs under some refinement q').

A rule satisfying the outcome-critical clause of N3 at some reachable position is called *decision-nondegenerate.*

Theorem 3 (Tier-3 exclusion under N1+N2)**.** *For any position x with $\mathrm{wt}(x) > c\,w_0$, the at most c queried functionals span only weight-$\le c\,w_0$ directions, so there exist refinements q, q' agreeing with every oracle answer but differing on $Q(x)$. Hence no in-class rule's move predicate factors through $Q(x)$ at dense positions – by theorem, not stipulation.*

Theorem 4 (clock completeness; machine-verified)**.** *N1+N2 alone, without N3, admit pure transport clocks in every semantics, including plain normal play: the pending-marker clock – positions*

(x, ε, i) *with invariant* $J = Q(x) \oplus \varepsilon$*,* $(1,1)$*-local, zero outcome-critical positions – satisfies N1+N2 exactly for every characteristic-*2 *form. Hence N3 is load-bearing: it carries the entire residual content of "natural".*

Remark 10 (N3 is stated to be attacked)**.** The *escape-edge* construction – a clock plus one dominated q-blind gadget edge from every nonzero position – passes N1, N2, and N3 (and a density-strengthened N3$^+$) while being morally a clock (`experiments/gold/skeptic_escape_edge_attack.py`). The natural repair, a dominance-pruned or two-game version of N3, runs into the fact that two-game criticality (critical in both the q- and q'-games with differing outcomes) is unsatisfiable in two-class outcome semantics. The anti-clock axiom therefore remains an open definitional problem; N3 is the best current formulation, recorded so that it can be attacked further.

Calibration (implemented and tested at the stated instances). Under F1–F3 + N1–N2: the T2 spin-flip rule of Section 10 fails N3 (zero outcome-critical positions, verified at $(8,1)$ and $(8,2)$); the pending-marker clock fails N3; ECHO-ko passes N3 (at least 160 mistake-states at each verified exact $m = 4$ instance); an order-4-alphabet loopy candidate from the run fails N2 (its alphabet is a 2^m-point Q-evaluation); an unwinding-game candidate from the run fails N1 (isotropic frames break exactness); and the repository's Tier-3 circuit and `interactive_kernel` Rule 2 fail N2 (dense-point form access). The candidates named here live in `experiments/gold/`.

10 Constructions and near-misses

10.1 T2: the width-2 spin-flip rule (exact, but a clock)

Proposition 4 (T2 realizes every char-2 quadric; implemented and tested)**.** *On positions* $x \in \mathbb{F}_2^n$*, let the rule be: turn over a set* d *of coins with* $\mathrm{wt}(d) \in \{1, 2\}$*, leading coin heads-to-tails, legal iff* $B(x, d) \oplus Q(d) = 1$*. Then for every characteristic-*2 *quadratic form* Q*, the normal-play* P*-set is exactly* $\{Q = 0\}$*.*

Proof sketch (blocking lemma). Every legal move flips Q (by the polarization identity, since $Q(x + d) = Q(x) + Q(d) + B(x, d)$ and legality forces $Q(d) + B(x, d) = 1$), so it suffices that every position with $Q(v) = 1$ has a legal descent. If all descents from v fail, then $B(v, e_i) = q_i$ for all $i \in \mathrm{supp}(v)$ and $B \equiv 0$ on $\mathrm{supp}(v) \times \mathrm{supp}(v)$; expanding $Q(v)$ in the basis then forces $Q(v) = 0$, a contradiction. □

Verification. Checked at $(8, 1)$, $(16, 1)$, the bent $(8, 1, \lambda = 2)$, and 20 random refinements of each B (the "refinement uniformity certificate"; `experiments/gold/witness_test.py` and companions). For $a = 1$ the rule constants q_i are game-native via Proposition 2; for $a \geq 2$ they are fed in, and game-nativity is open.

T2 is, however, a forced clock: every legal move flips Q, play telescopes $Q(x)$ as a parity count, every position's options share one outcome class, and no position has a losing move. By Theorem 1 this is unavoidable for *all* in-class normal-play realizers, not a defect of T2 specifically. T2 and the pending-marker clock pin the boundary: within every tested bounded-access quarantine, exact realizers exist but are forced clocks.

10.2 echo-ko: charge counting on the extraspecial cocycle

The one known exact realizer that is *not* a clock works one level up, on the extraspecial cocycle itself.

Rule (echo-ko). Positions are indexed by $x \in \mathbb{F}_{2^m}$; the coins of x must each be touched twice; players alternate single touches; each touch of e_i updates a charge $\sigma \oplus= c(\text{open-set}, e_i)$, where c is the triangular cocycle of the extraspecial extension ($c(v,v) = Q_a(v)$, $c(u,v) \oplus c(v,u) = B(u,v)$, cf. Lemma 1(ii)); a one-move ko forbids immediately re-touching, a player with no legal touch passes (the pass clears the ko), and play ends when every coin is closed. The two players are the two central characters: the complete play word lies over $0 \in V$, hence equals 1 or z in E_Q; one player plays for terminal charge $\sigma = 1$ (word $= z$), the other for $\sigma = 0$, and the *value* of the position is the forced terminal charge.

The rule is center-reading (not center-blind), noncommutative via move order (not turn count), and – at every verified exact instance – decision-nondegenerate. For $a = 1$ its constants are game-built via Proposition 2, with no per-instance Q-evaluation.

Provenance and results (implemented and tested). A first-round solver had a bug (the memoization key omitted the accumulated charge parity); the corrected solver was validated against explicit tree enumeration. Corrected results (`experiments/gold/echo_charge_probe.py` and companions):

instance	core	agreement	decision-live
$m = 4$, bent $\lambda \in \{2, 12\}$	rank 4	16/16 exact	yes (≥ 160 mistake-states)
$(8, 2)$, $\lambda = 1$	rank 4	255/256 (miss $x = 224$)	yes
$(8, 1)$, $\lambda = 1$	rank 6	228/256	yes
$(8, 1)$, bent $\lambda = 2$	rank 8	212/256	yes

Agreement grades by polar rank. The table reports the σ=1-stance face (first player plays for z) of a stance-asymmetric rule: with the first player playing for $\sigma = 0$ the same instances degrade (15/16 at the bent $m = 4$ instances, 250/256 at $(8, 2, 1)$). An independent σ-explicit full-state direct solver, validated against tree enumeration (`experiments/echo_solver.py`, stage `ko2`; 2026-06-10), reproduces the table exactly, including the unique $(8, 2)$ miss $x = 224$ – so the corrected round-1 numbers now have an independent re-derivation. The FIFO+dummy variant below is, by contrast, exact at *both* stances. A bounded-memory blocker conjecture (for fixed ko-window w, adversarial unlinking wins on supports $k \geq w+2$ at rank ≥ 6) was formulated on the *invalid* round-1 data and has not been re-examined against the corrected solver; it is recorded as a hypothesis to re-test, not a finding.

10.3 echo-fifo+dummy: the verified exact $m = 8$ realizer

A second-round variant imposes a FIFO close-discipline on top of the rule above – only the longest-open coin may be closed, while the one-move ko and the forced pass are *retained* – and adjoins one neutral tempo coin ($q = 0$, zero polar row) to every board, fully queued and ko-able. The readout is σ-valued as in ECHO-ko, at both stances $t \in \{0, 1\}$; "exact" means the forced terminal charge equals $Q(x)$ for every position and both stances (765 scaled forms $\times$ 256 positions $\times$ 2 stances = 391,680 checks). The original run established this via a decomposition-plus-isomorphism-caching solver validated only at $m = 4$, and earlier drafts recorded the claim as unverified.

Adversarial review (2026-06-10; implemented and tested). The pre-registered experiment of Section 10.4 was executed with a fresh direct stateful solver (`experiments/echo_solver.py`):

no decomposition, no isomorphism caching, full state – untouched set, open queue in order, ko memory, mover, and the accumulated charge σ – in the memoization key. Validation: explicit no-memo tree enumeration ($m = 4$ exhaustive, $m = 8$ small supports); the original direct $m \leq 4$ solver executed verbatim as a cross-oracle (1,920 solves agree); nim arithmetic against the independent Turning-Corners mex recurrence; and a second reviewer model re-ran every stage and re-checked the nim product against the original probe's implementation on all 65,536 pairs. Results:

- $m = 4$ family: exact at all 15 scaled forms, both stances, with the dummy (30/30; only 15/30 without it – the tempo coin is load-bearing).
- $m = 8$ benchmarks $(8, 2, 1)$ rank 4, $(8, 1, 1)$ rank 6, $(8, 1, 2)$ and $(8, 1, 3)$ rank 8: 256/256 at both stances.
- Rank-stratified sample (21 (a, λ) pairs across ranks $\{4, 6, 8\}$): exact, both stances.
- The full claimed sweep, re-derived end to end: **391,680/391,680**, zero misses.
- Decision-liveness: 1.46–4.37 million decision states per benchmark instance (128–210 of 256 positions decision-live) – not a clock.
- Torsor sweep: 20 refinements of each benchmark polar form (Gold, zero, all-ones, seeded-random diagonals), exact throughout. Honest note: each q_i fires exactly once, at the close of coin i, so the diagonal contribution to σ is play-invariant; both-stance exactness at one refinement of B therefore *implies* it for all refinements, and the sweep is a consistency check of that argument, not independent evidence.

The exactness claim is hereby CONFIRMed. Two boundary notes. First, the verified realizer is σ-valued: it realizes Q as a forced terminal charge (the central character of the play word), not yet as the P-set of a normal-, misère-, or loopy-play game; the recasting is now the load-bearing open step (Section 11). Second, the bounded-memory blocker conjecture is untouched: the FIFO queue is unbounded memory, so the bounded-window regime remains open – what the verdict shows is that unbounded queue discipline plus one tempo coin suffices.

10.4 The decisive experiment

The pre-registered experiment for the primary candidate: a corrected-solver sweep over the finite ECHO rule family at $m = 8$, validated by a direct stateful solver.

- *Benchmarks*: $(8, 1, \lambda = 1)$ rank 6; bent $(8, 1, \lambda = 2)$ rank 8; $(8, 2, \lambda = 1)$ rank 4; the full $m = 4$ family as regression.
- *Axes*: ko-memory window $w \in \{1, 2, 3\}$; pass semantics (clears-ko / forbidden / loses); single-coin plus pair touches (the tartan-companion axis); both orientations.
- *Success criterion*: zero misses across all tested (m, a, λ) triples, decision-nondegenerate throughout, and surviving a torsor sweep (≥ 20 stratified refinements of each B, confirming refinement uniformity rather than single-refinement luck).

A CONFIRM outcome would be the first genuine Tier-2 witness (pending recasting the charge readout into normal/misère/loopy semantics and the even-a diagonal lemma). A KILL outcome – rank-graded decay persisting across every family member – would put the bounded-memory blocker conjecture on valid data and close the line for bounded-memory architectures.

Executed 2026-06-10 (Section 10.3): outcome CONFIRM on every leg – zero misses, decision-nondegenerate throughout, torsor-uniform. The remaining family axes (ko-memory window $w \in$

$\{1, 2, 3\}$, pass semantics, pair touches) are no longer decisive for existence; they map the boundary of the mechanism – which disciplines besides FIFO+dummy are exact, and why – and feed the reconciliation item of Section 11.

10.5 The linking reduction and the general-m theorem

The mechanism question of the verified realizer reduces, by three short play identities, to one combinatorial statement about graphs (`experiments/linking_game.py`; second-pass run 2026-06-10). Fix a board: coins = vertices of a graph G (on a Gold board, G is the polar graph B restricted to $\mathrm{supp}(x)$, plus the dummy as an isolated vertex; the diagonal q is play-invariant and splits off, as in the torsor note of Section 10.3).

Reduction (standard math; each step is a whole-play identity, machine-validated on random plays).

1. *No nesting.* FIFO forces coins to close in opening order, so no open-window nests inside another; an edge of G is linked iff its two windows *overlap*, and σ equals the parity of $E(G)$-edges covered by the play's interval-overlap graph.
2. *Odd-close accounting.* Charging an edge when its second endpoint opens (same play total), set $\mathrm{und}(s)$ = the number of edges with an untouched endpoint. Then $D = \sigma \oplus \mathrm{und}$ is invariant under opens and passes, and flips exactly when the queue front f closes with $\deg_U(f)$ odd (U = untouched set). The whole game is therefore the *odd-close parity game*: D starts at $|E| \bmod 2$, only odd-degree front closes flip it, and $\sigma_{\mathrm{final}} = D$.
3. *Ko localization.* Opens of untouched coins are never ko-blocked; a close is ko-blocked only when the front was opened onto an empty queue on the immediately preceding move; forced passes occur only once $U = \emptyset$, after which no flip is possible. Passes are irrelevant to the flip fight.

The linking theorem (target). *If the board contains an isolated coin, the flip count is forced even – from both seats, for every graph.* Equivalently σ is forced to $|E| \bmod 2$, which on a Gold board is $Q(x)$: m-uniform exactness of ECHO-FIFO+dummy.

Verified (implemented and tested). The statement holds for *every* isomorphism class on $k \leq 7$ real coins plus dummy, both seats (1,044 classes at $k = 7$) – strictly stronger ground than the Gold-arising boards of the $m = 8$ sweep. Without the isolated coin the failures ("Bad graphs", census 1/4/34 at $n = 3/5/7$, none at even n) are always *mover-controlled* – the first player forces whichever parity they want, never the anti-parity – and none contains an isolated vertex; 33/34 at $n = 7$ have a dominating vertex.

Device lemma (standard math). With the queue empty, if some $v \in U$ dominates $R = U \setminus \{v\}$ with $|R|$ even and nonempty, the mover forces a flip in two plies: open v (ko-protected, $\deg_U(v)$ even), every reply must open some $y \in R \subseteq N(v)$ making v odd, close v. Conversely, a single odd front at the responder's turn is always re-evenable (odd degree ≥ 1 supplies a never-ko-blocked neighbor open), and when v does not dominate R a non-neighbor open preserves evenness: the domination-with-even-remainder pattern is the unique *local* obstruction. An isolated coin defeats it at every root (nothing dominates a set containing it), matching the Bad census; for $|R|$ odd, domination makes deg odd and the neighbor reply works, explaining the bonus rigidity of all even-n boards.

A verified defender strategy. A two-mode menu strategy tracking the flip debt g – PREVENTION ($g = 0$: re-even odd fronts; non-toggling opens, e.g. the dummy; safe closes; in the trap $U \subseteq N(\text{front})$, poison or close) and DEBT ($g = 1$: counter-close odd fronts – the attacker's flip-close typically exposes the repair; else toggle or advance) – beats an optimal unrestricted attacker on every class $k \leq 7$, both seats, with no move outside the menus. The verification is menu-*existential*: the menus always contain a winning move, but not every menu choice wins (a losing poison choice on the star was exhibited in review).

Open residue. The general-n proof. Parity-local invariants provably do not suffice: over 13 natural parity features of the state, the safe/unsafe label is not a function of the features, and minimal distinguishing pairs differ precisely in whether the untouched structure retains a repair device – a recursive, not parity-local, condition. The working architecture (second-eyes review concurring): segment the queue at *firewall* coins ($\deg_U \equiv 0$; the opened dummy is a permanent firewall, the untouched dummy a virtual one), and run a mutual induction – no-debt and one-debt regimes – per segment, with attacker constructions classified by certificate depth (leaves exactly the device lemma's obstruction) rather than by graph shape. The hard obligation is the poison transition: showing every reachable trap state offers a move into a segment with repair potential.

11 Status and next moves

Status of the trichotomy. Tier 1 ($\mathrm{Sp}(B)$-blind) is dead by Proposition 1 and Theorems D–G. Tier 3 (evaluation) is live and even decision-nondegenerate, but tautological. Tier 2 under the bounded-framing quarantine is inhabited *only by clocks* among normal-play realizers (Theorems 1, 4). In σ-valued (forced-charge) semantics the picture changed on 2026-06-10: the ECHO-FIFO+dummy realizer is *verified* exact on all 765 scaled $m = 8$ Gold forms at both stances (Section 10.3), decision-nondegenerate in bulk, with bounded per-move access (open-set row sums plus the local q_i at closes) – an N1+N2-satisfying non-clock, in a readout the rigidity theorems do not cover. The mechanism question was then reduced (Section 10.5) to the linking theorem – flips forced even on any board with an isolated coin – now verified for every graph class through $k = 7$ with an explicit two-mode defender strategy; what remains of it is the general-n induction. The live questions are *semantics* (recast the charge readout into normal/misère/loopy outcome classes, or prove the recasting impossible), the general-n linking proof, and the even-a diagonal lemma.

Ranked next moves. (Move 1 of the previous ranking – the adversarial review – was executed 2026-06-10 with outcome CONFIRM, Section 10.3; the mechanism-reconciliation half of the old move 1 was then executed in the same-day second pass, Section 10.5: the reduction is done, the mechanism is the odd-close parity game, and FIFO's role – no nesting, hence overlap = linking – and the dummy's role – killing the domination device at every root – are identified.)

1. *Recasting* (highest leverage): recast the σ-valued charge readout into normal/misère/loopy outcome semantics – or prove the recasting impossible.

1′. *The general-n linking proof*: close the open residue of Section 10.5 (firewall segmentation, mutual no-debt / one-debt induction, certificate-depth completeness); this upgrades m-by-m verification to the theorem for all m.

2. *Family-boundary sweep* (the remaining pre-registered axes): ko-memory window $w \in \{1, 2, 3\}$, pass semantics, pair touches, no-dummy controls – map which disciplines are exact; this also puts the bounded-window blocker conjecture on valid data at last.

3. *The even-a diagonal lemma*: prove or refute the even-a $\wp$-preimage family (Section 6.3) – the second condition on the Tier-2-witness reading of the verified realizer.

4. *Frobenius-aware access enumeration*: finite enumeration of Frobenius-equivariant legality predicates at $m = 4, 8$ – the one access window (escape hatch 3 of Remark 9) where both the symmetry and oracle methods are provably silent.

5. *Cheap gates*: (a) verify the regularity hypothesis of [12, Thm. 6.4] against the published paper (load-bearing for Theorem C); (b) enumerate conjugation-move rules on E (the involution locus is conjugacy-invariant, so Theorem H's left-translation kill does not apply); (c) exhaust the board-8 case of the FIFO parity-pinning conjecture.

12 Validation

The claims above are backed by implementation checks, not by a finished proof of the open game statement.

- `cargo test` exercises the scalar arithmetic, Clifford product, Arf computation, quadratic fitting, misere machinery, and related invariant modules.
- `experiments/trace_form_arf.py` checks the Gold rank formula (Table 1).
- `experiments/gold_form_from_games.py` and `experiments/tartan_bilinear.py` rebuild the Gold form and polar form from literal Turning-Corners products in small fields.
- `experiments/arf_win_bias.py` checks the zero-count formula (Table 2).
- The `misere_quotient`, `octal_hunt`, `interactive_kernel`, `loopy_quadric`, and `bent_route` examples implement the route probes of Section 6.1.
- `experiments/gold_family_survey.py` checks the scaled-component bent counts; `experiments/framing_obstruction.py` the diagonal-framing pattern; `experiments/misere_kernel.py` the R_8 kernel reading.
- `experiments/gold/` carries the probes of the 2026-06-10 parallel run backing Sections 8–10: exhaustive symmetry checks at $m = 4$, the B-oracle and flip-rule kills, the extraspecial and ECHO solvers, and the criterion calibration. These are machine-generated research probes, run under the project virtualenv, and are *not* CI-maintained; each result cited from them states its verification scope inline above.
- `experiments/echo_solver.py` is the adversarial-review harness of the 2026-06-10 verification pass (stages `selftest`, `ko2`, `fifo2-validate` through `fifo2-all`): direct full-state solvers for the ECHO family, self-tested against the Turning-Corners mex recurrence, explicit tree enumeration, and the original direct solver run verbatim. Unlike `experiments/gold/` it is dependency-free (stdlib Python) and written to stay reproducible.
- `experiments/linking_game.py` is the linking-reduction harness of the second 2026-06-10 pass (stages `validate`, `screen`, `strategy`; Section 10.5): the abstract odd-close parity game, the reduction identities on random plays, the all-classes $k \leq 7$ rigidity and Bad-graph screens, and the strict menu verification of the two-mode defender strategy. Stdlib-only; cross-validated against `experiments/echo_solver.py` through its `SynthForm` bridge.

Two probe-hygiene caveats from the run remain open in the repository and are recorded in Remark 8: the `octal_hunt` target should be reframed (its group-quotient target is empty under Theorem C), and `p_set_as_f2` should verify that its subset labeling is a homomorphism.

13 Conclusion

The meaningful content is layered. At the base: game-operation-built characteristic-2 trace forms inside a nimber Clifford backend, with a precise conditional interpretation of Arf as a P-set bias. Above it: the linear ceiling (Theorem A, with the lexicodes as its rich solved floor), the extraspecial identification of the missing quadratic datum with a noncommutative central extension (Lemmas 1–3), and a no-go program (Theorems B–H, 1–2) that empties every symmetric, commutative, and oracle-bounded architecture of bulk strategic content. At the boundary: exact normal-play realizers exist in every tested quarantine class but are forced clocks; ECHO-ko is exact on the $m = 4$ bent instances at one stance, degrades gracefully with polar rank, and stays decision-live; and the ECHO-FIFO+dummy realizer is now *verified* exact on all 765 scaled $m = 8$ Gold forms at both stances – decision-nondegenerate, bounded-access, and σ-valued (Section 10.3). A formalized criterion (N1–N3) reduces "natural" to one load-bearing, still-attackable axiom. The draft does not solve the game-semantics problem; the verified realizer narrows it to a named question: *can the forced-charge readout be recast into normal, misère, or loopy outcome semantics on a Gold quadric of core rank* ≥ 6 *– or is the two-class outcome layer itself the obstruction?*

Acknowledgements

The `ogdoad` library, the experiments, and this draft were developed with LLM assistance; Sections 8–11 synthesize the output of a structured parallel research run (2026-06-10) whose probes are committed under `experiments/gold/`. Results from that run are reported with their verification scope, and one construction claim is explicitly flagged unverified. Mathematical direction, framing, and responsibility for claims remain the author's.

References

[1] C. Arf, *Untersuchungen über quadratische Formen in Körpern der Charakteristik 2, I*, J. Reine Angew. Math. **183** (1941), 148–167.

[2] C. L. Bouton, *Nim, a game with a complete mathematical theory*, Ann. of Math. **3** (1901/02), 35–39.

[3] E. R. Berlekamp, J. H. Conway, R. K. Guy, *Winning Ways for Your Mathematical Plays*, 2nd ed., A K Peters, 2001–2004.

[4] J. H. Conway, *On Numbers and Games*, Academic Press, 1976.

[5] J. H. Conway, N. J. A. Sloane, *Lexicographic codes: error-correcting codes from game theory*, IEEE Trans. Inform. Theory **32** (1986), 337–348.

[6] L. E. Dickson, *Linear Groups, with an Exposition of the Galois Field Theory*, Teubner, 1901.

[7] R. Elman, N. Karpenko, A. Merkurjev, *The Algebraic and Geometric Theory of Quadratic Forms*, AMS Colloquium Publications **56**, American Mathematical Society, 2008.

[8] R. Gold, *Maximal recursive sequences with 3-valued recursive cross-correlation functions*, IEEE Trans. Inform. Theory **14** (1968), 154–156.

[9] D. Gorenstein, *Finite Groups*, 2nd ed., Chelsea Publishing, New York, 1980.

[10] R. Lidl, H. Niederreiter, *Finite Fields*, 2nd ed., Cambridge University Press, 1997.

[11] V. Ovsienko, *Real Clifford algebras and quadratic forms over $\mathbb{F}_2$: two old problems become one*, The Mathematical Intelligencer **38** (2016); arXiv:1601.07664.

[12] T. E. Plambeck, A. N. Siegel, *Misere quotients for impartial games*, J. Combin. Theory Ser. A **115** (2008), 593–622.

[13] D. Quillen, *The mod 2 cohomology rings of extra-special 2-groups and the spectra of quadratic forms*, Math. Ann. **194** (1971), 197–212.

[14] D. E. Taylor, *The Geometry of the Classical Groups*, Sigma Series in Pure Mathematics **9**, Heldermann Verlag, 1992.

[15] D. L. Winter, *The automorphism group of an extraspecial p-group*, Rocky Mountain J. Math. **2** (1972), 159–168.